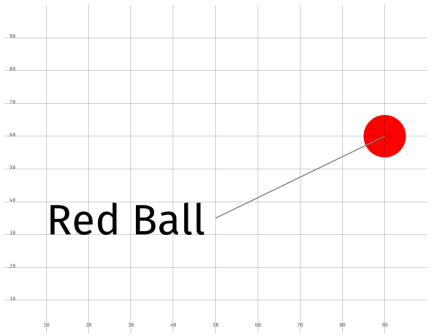
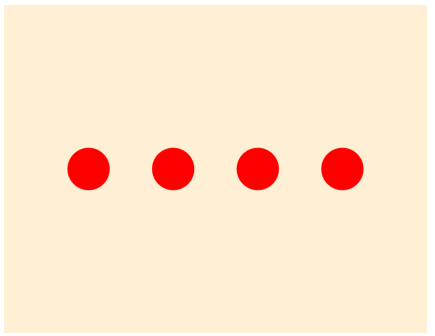
The decksh Tutorial

Your First Deck

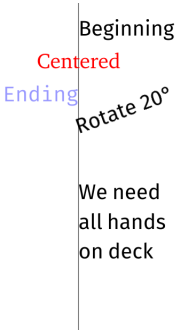


The Coordinate System

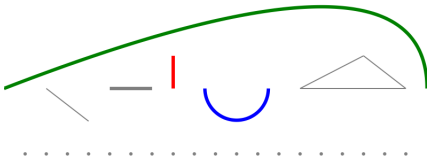
Variables



Using Color



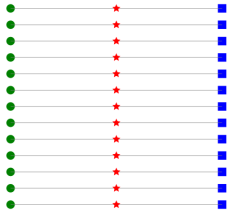
Working with Text



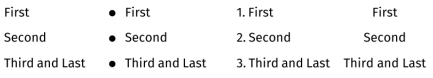
Stroked Graphics

Filled Shapes

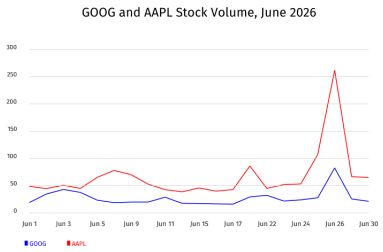
Images



Loops



Lists



Charts

Geographic Functions

Your First Deck

This is your first deck. Congratulations.

A deck is usually contained in a single text file, conventionally with the suffix ".dsh". Each line in a deck begins with a keyword: a single word that reflects its function. Comments begin with '//' and may be placed at the beginning, or anywhere on a line of decksh code.

The code for the deck is contained between the 'deck' and 'edeck' keywords. The 'slide' and 'eslide' keywords define the boundaries of a slide. You may include an arbitrary number of slides in a single deck.

Keywords are followed by items, known as 'arguments', separated by spaces or tabs, that define the keyword's behavior. In the example, the 'slide' keyword has two arguments: "black" and "white", which means the slide background is black and its text is white. If you omit the optional arguments, the default is black text on a white background.

The 'ctext' keyword means 'centered text'. Its arguments are the text ("hello, world"), followed by the coordinates of the text, and followed by the size of the text. The image is placed at the bottom of the slide, scaled to span the slide's width.


```
// First
deck
    slide "black" "white"             // black background, white text
        ctext "hello, world" 50 50 15 // say it loud
        image "earth.jpg"    50 0 100 0 // image at the bottom
    eslide
edeck
```

Think of a deck as a series of pages; the building blocks of your story.

The Coordinate System

Decksh uses the traditional Cartesian coordinate system to place elements on the canvas. The x, y coordinates represent percentages ranging from 0-100; the coordinates are passed as arguments as a pair of numbers, x first, then y.

The origin (0,0), is at the lower-left, with the x coordinate increasing from left to right, with the y coordinate increasing bottom to top. The middle of the canvas is (50,50): half-way from the left and half-way up from the bottom.

Other measures like the size of text or other objects are represented by the percent of the side width.

For example, to place the text "Red Ball" left-aligned at 10% from the left, and 30% up (10,30), and the text size at 10% of the canvas width, use:

text "Red Ball" 10 30 10

Other elements like circles and lines may also be placed. For example:

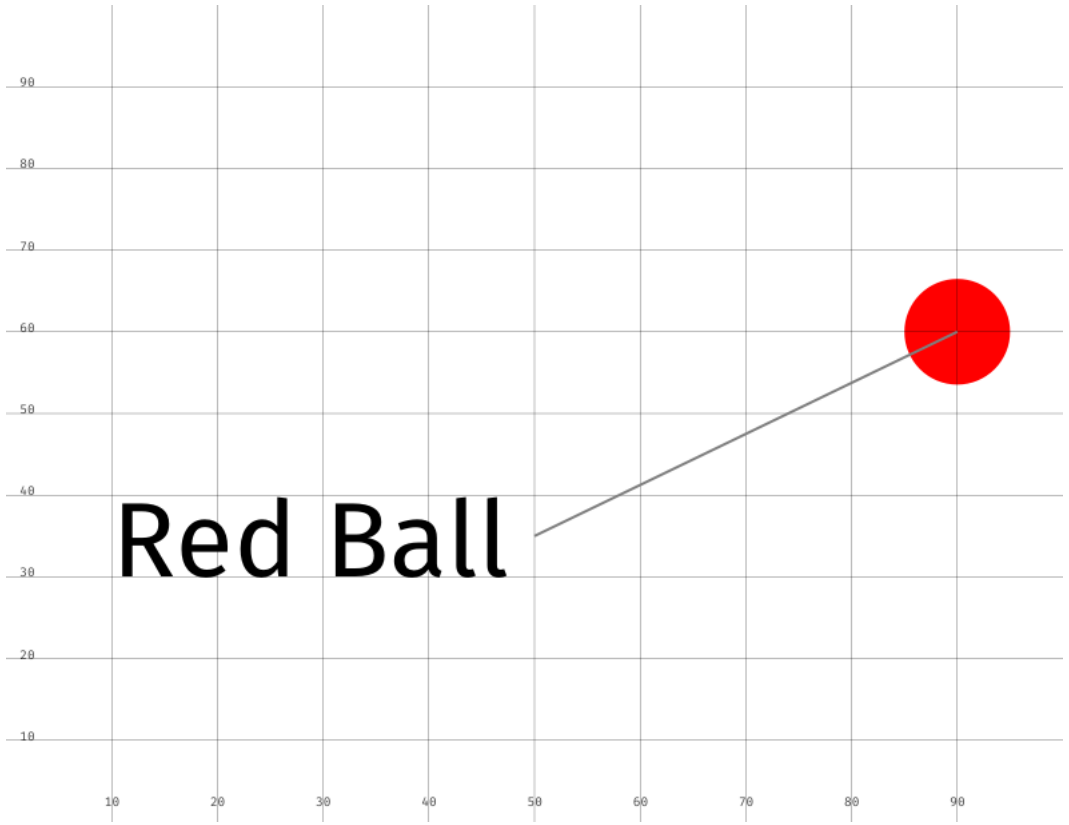
circle 90 60 10 "red"

places a red circle at (90,60), the size of the circle is 10% of the canvas width.

To draw a line from (50,35) to (90,60):

line 50 35 90 60

Finally, note the 'ruler' keyword, which makes a convenient scale.




```
deck
    slide
        ruler 10
        text "Red Ball" 10 30 10
        circle 90 60 10 "red"
        line 50 35 90 60
    eslide
edeck
```

`Place anything, anywhere with precision and consistency.`

Variables

Variables provide a method for attaching a name to an object so that the name may be used for other operations.

Variables may refer to either numbers, or text strings (data defined between " characters). Use the '=' after the name to assign the value of variable. The general syntax is:

name = value

In the example, (1) the value of "hello" is assigned to s, (2) 5 and 50 to the variables x and y and (3) ts is assigned the value of 5. Next, use the variables:

text s x y ts

which is equivalent to:

text "hello" 5 50 5

When using variables that contain numbers, you can also perform binary operations: add (+), subtract (-), multiply (*), divide (/), and modulo (%). For example,

a = x + y // a gets the sum of x and y

You can also use 'assignment operations' on a single variable, for example to add 10 to an existing variable x, use:

x += 10


```
deck
    slide
        s="hello"
        x=5
        y=50
        ts=5
        text s x y ts
        x=x+30
        ts+=15
        text s x y ts
    eslide
edeck
```

`Use variables as the levers that control the deck's elements and their relationships.`

Using Color

Many decksh elements use color as an optional argument.

The color may be specified in four ways:

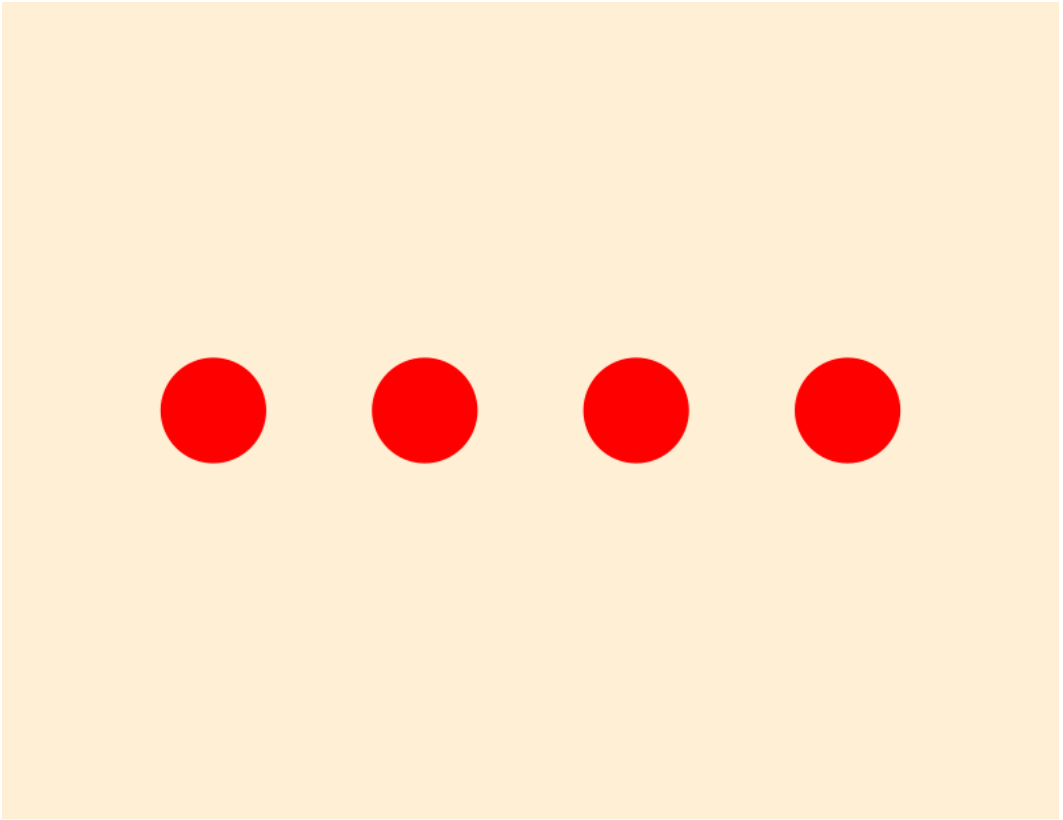
(1) A named color as defined in the SVG standard; these are colors like "red" and "blue", but also include fairly exotic names like "papayawhip".

(2) A RGB triple, which uses numbers ranging from 0-255 to refer to amount of red, green, and blue in the color. For example, "rgb(0,0,0)" is black, "rgb(255,255,255)" is white, with "rgb(128,128,128)" as gray. By the way, papayawhip is "rgb(255,239,213)".

(3) The hex RGB format, in the form of "#rrggbb", where rr, gg, and bb are the red, green and blue components of the color as hexadecimal digits ranging from "00" (zero) to "ff" (255). Papayawhip is "#ffefd5" in hex.

(4) The HSV (hue, saturation, value) format uses three numbers: the first, hue, ranges from 0-360; the saturation and value (brightness) range from 0-100.

For example, "hsv(0,100,100)" is a fully saturated red, and "hsv(120,50,50)" green with diminished saturation and brightness.



```
deck
    slide "papayawhip"
        c1="red"            // named color
        c2="rgb(255,0,0)"   // rgb
        c3="#ff0000"        // hex rgb
        c4="hsv(0,100,100)" // hue, saturation, value
        circle 20 50 10 c1
        circle 40 50 10 c2
        circle 60 50 10 c3
        circle 80 50 10 c4
    eslide
edeck
```

Use color as a method of communicating meaning, regardless of how you name it.

Working with Text

Text is an important decksh element and may be used in various ways. Most text keywords use this pattern:

keyword "text" x y size [font] [color] [op]

Text may be aligned (begin, end, and centered) relative to a specific point. The corresponding keywords are 'text' (and its synonym 'btext') 'etext' and 'ctext'. The point is specified by the x and y coordinate, the size of the text is the third 'size' argument (a percentage relative to the width of the canvas)

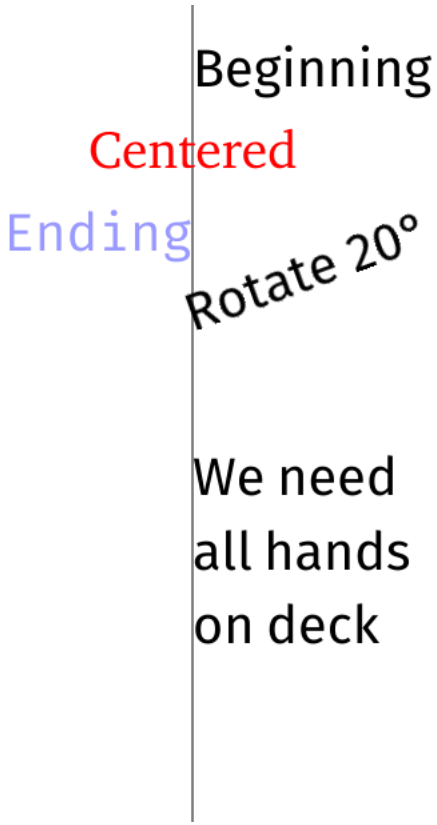
All text keywords have an optional set of arguments that define the font (either "sans", "serif", "mono", or "symbol"), the color of the text, and its opacity (ranging from 0 (invisible) to 100 (solid color)) If no optional arguments are specified the default values are "sans", "black", and 100.

Rotated text ('rtext') is specified like the others, except you must include the angle of rotation (0-360).

rtext "text" x y angle size [font] [color] [op]

A block of text is also similar, but the width of the block must be specified.

textblock "text" x y width size [font] [color] [op]




```
deck
    slide
        tx=50
        ts=5
        saying="We need all hands on deck"
        line                  tx 0 tx 100
        btext "Beginning"     tx 90    ts "sans"
        ctext "Centered"      tx 80    ts "serif" "red"
        etext "Ending"        tx 70    ts "mono" "blue" 40
        rtext "Rotate 20°"    tx 60 20 ts
        textblock saying      tx 40 20 ts
    eslide
edeck
```

Build your story using text.

Stroked Graphics

decksh supports stroked graphic shapes including lines, polylines, arcs, and curves.

The general syntax includes the arguments for the shape, with optional arguments for line width, color and opacity; (defaulting to 0.2%, "gray" and 100%)

To draw a line between (10,50) and (20,70):

line 10 50 20 70

Dotted lines are similar, but you may specify the gap between dots after the line width:

dline 10 50 90 50 0.2 5

Horizontal and vertical lines specify the beginning point and the length:

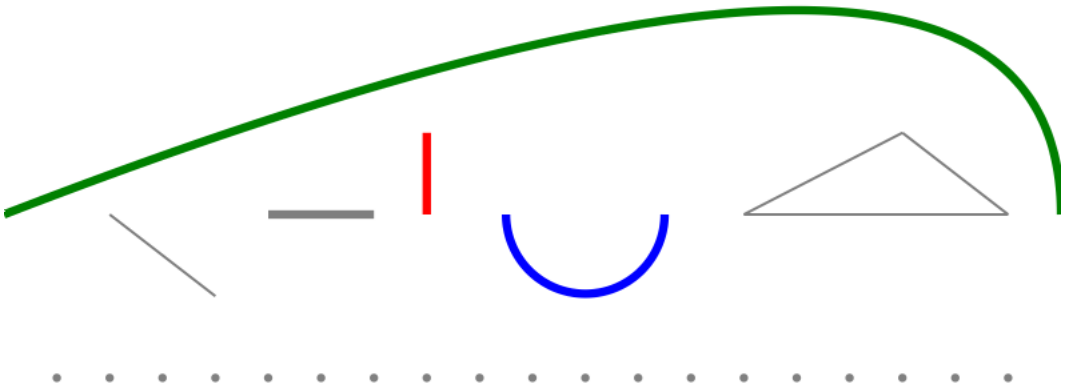
hline 10 50 30 and vline 10 50 30

For arcs, specify the center of the arc, its width and height, and start and ending angles.

arc 50 50 10 10 0 90

Polygons use two string arguments: the first is a list of x coordinates, and the second is the y coordinates.

Quadradic Bezier curves take three pairs of coordinates: the beginning point, the control point, and the end point.



```
deck
    slide
        base=50
        lw=0.8
        line     10 base 20 40
        hline    25 base 10 lw
        vline    40 base 10 lw "red"
        arc      55 base 15 15 180 360 lw "blue"
        polyline "70 85 95" "base 60 base"
        curve     0 base 100 100 100 base lw "green"
        dline     5 30 95 30 lw 5
    eslide
edeck
```

You have to draw the line somewhere.

Filled Shapes

decksh supports filled shapes including circles, ellipsis, square, rectangle, rounded rectangle, star, and polygon.

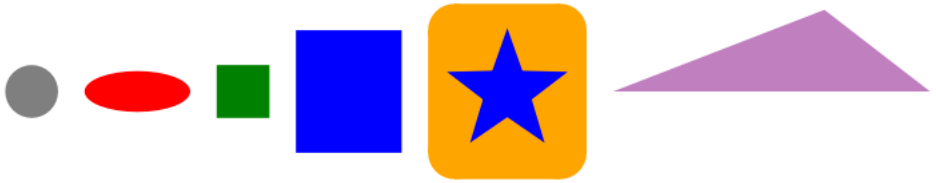
Filled shapes are specified by location arguments (x and y coordinates usually at the center of the shape), followed by dimensions (width and height), with optional arguments for color and opacity; if optional arguments are omitted, the color is "gray" with 100% opacity.

Circles and squares take the center point and a single dimension (width). Ellipses and rectangles also take a center point with a width and height.

Rounded rectangles are specified like regular rectangles, but they require you to specify the radius of the corners.

Filled polygons are specified like polylines: two string arguments that specify the sets of x and y coordinates.

To make stars, specify the center of the star, the number of points, and the inner and outer radii.



```
deck
    slide
        base=50
        circle    10 base 5
        ellipse   20 base 10 5              "red"
        square    30 base 5                 "green"
        rect      40 base 10 15             "blue"
        rrect     55 base 10 15 5           "orange"
        star      55 base  5 2 6            "blue"
        polygon   "65 85 95" "base 60 base"  "purple" 50
    eslide
edeck
```

Use these elements to shape your slides.

Images

An image is the welcome companion to text and graphics when making decks.

To place an image, you specify its name (JPEG, PNG, and GIF image formats are supported), location, and dimensions. The general syntax is:

image "name" x y width height

The specified coordinate refers to the middle of the image, and the width and height refer to its dimensions (in pixels). An alternative (and recommended) syntax scales the image to the width of the canvas:

image "name" x y canvas-width 0

For example:

image "follow.jpg" 50 50 30 0

Places the JPEG image in the middle of the slide, scaled at 30% of the canvas width.

To add an image caption, use the 'cimage' keyword. The syntax is similar, but an additional caption argument is used:

cimage "cloudy.jpg" "Cloudy Day" 50 50 50 0

Head in the clouds

```
deck
    slide
        image  "follow.jpg"                       25 50 45 0
        cimage "cloudy.jpg" "Head in the clouds" 75 50 45 0
    eslide
edeck
```

```
The artful arrangement of pictures help
illuminate your deck.
```

Loops

Loops allow you repeat decksh operations in a contolled way.

A loop is defined as statements bounded by the 'for' and 'efor' keywords. The 'for' keyword arguments define the beginning, ending, and optional increment. This defines how many times the statements between 'for' and 'efor' are executed. The variable specified in the 'for' statement is substituted in the these statements. For example:

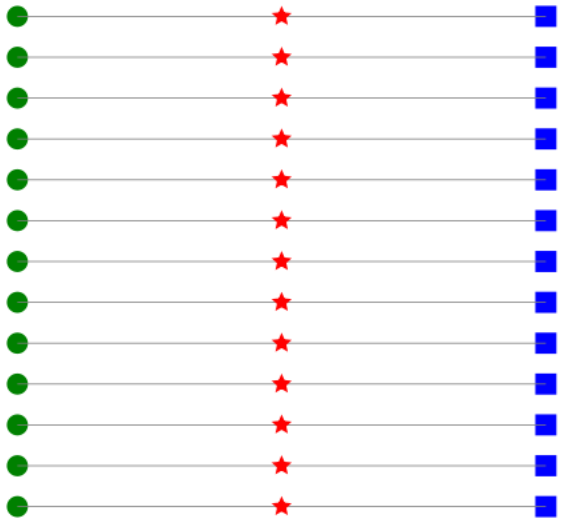
for v=10 50

assigns the value 'v' from 10 to 50, with an increment of 1.

for v=10 50 10

counts from 10 to 50 by 10 (10, 20, 30, 40, 50).

In the example on the right, a loop assigns 'v' from 10 to 80 by 5, and draws horizontal lines, circles stars, and squares using the value of v.



```
deck
    slide
        for v=20 80 5
            hline  20 v 50 0.1
            circle 20 v 2        "green"
            star   45 v 5 1 0.4  "red"
            square 70 v 2        "blue"
        efor
    eslide
edeck
```

```
'Round and 'round we go. I'll stop when you
tell me so.
```

Lists

With decksh you can make many kinds of lists: plain, bulleted, numbered, centered.

The type of list is specified by the keyword: 'list' for plain, 'blist' for bulleted, 'nlist' for numbered, 'clist' for centered.

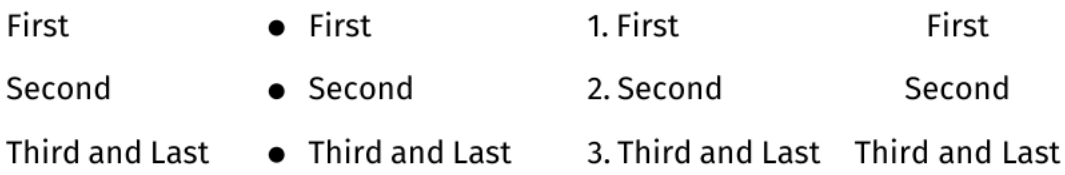
The arguments for lists include the x and y location and the text size of the list.

For example to begin a list at (20, 40) with text size 3 use:

list 20 40 3

Following the list type is any number of list items (li "item name"), one per line. The list ends with "elist" on a line by itself.

You can construct tables using "parallel lists"; aligning a series of plain lists to a set of columns.




```
deck
    slide
        ls=3
        ly=95
        list 0 ly ls
            li "First"
            li "Second"
            li "Third and Last"
        elist
        blist 25 ly ls
            li "First"
            li "Second"
            li "Third and Last"
        elist
        nlist 55 ly ls
            li "First"
            li "Second"
            li "Third and Last"
        elist
        clist 90 ly ls
            li "First"
            li "Second"
            li "Third and Last"
        elist
    eslide
edeck
```

Make a list and check it twice.

Charts

decksh has built-in support for a variety of chart types with these corresponding keywords: barchart, linechart, areachart, scatterchart and dotchart.

To make a chart, specify the chart type and the input file. Data files contain a series of tab separated name,value pairs. For example, a line of data looks like: January<tab>100

Optional arguments control the chart data, label, and value colors. For example to make a line chart from data in "foo.d":

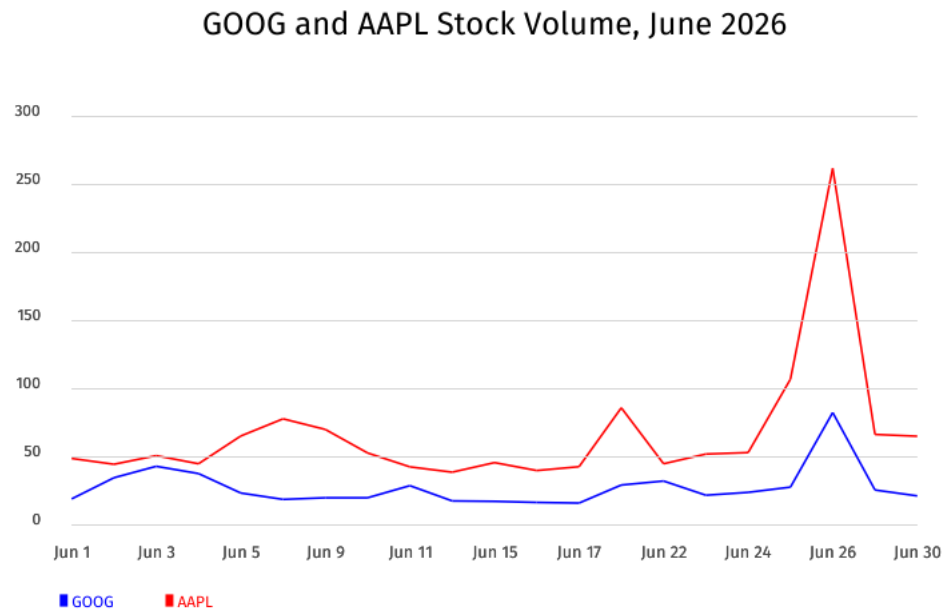
linechart "foo.d" [color] [label-color] [value-color]

Special variable names control the placement and other aspects of the chart. The variables 'chartTop', 'chartBottom', 'chartLeft', and 'chartRight' define the chart boundaries. If these are not set, the default values are: chartTop=80, chartBottom=30, chartLeft=10, and chartRight=90.

Other variables control items the visibility of values (chartVal), text size (charTextSize) and X-axis label interval (chartXLabel).

Chart legends are set with the 'legend' keyword using this syntax:

legend "text" x y size [font] [color]




```
deck
    slide
        ctext "GOOG and AAPL Stock Volume, June 2026" 50 90 3
        chartVal=0
        chartTextSize=2
        chartXLabel=2
        chartYRange="0,300,50"
        linechart "aapl-vol.d" "red"
        linechart "goog-vol.d" "blue"
        legend    "GOOG" 10 20 1.5 "sans" "blue"
        legend    "AAPL" 20 20 1.5 "sans" "red"
    eslide
edeck
```

Chart your course.

Geographic Functions

decksh can read shapefiles, KML, or geoJSON encoded data to produce geometric regions, borders and paths. Plain and labeled locations are also supported.

The keywords for geo functions include 'geoborder' (geographic borders), 'georegion' (regions; useful for countries and continents), 'geopath' (paths between locations), 'geolabel' (a simple label at a location), 'geoloc' (aligned labels ("begin", "center", "end") at a location), and 'geomark' (placing a dot at a location).

To place a map using GIS data in KML format:

georegion "africa.kml"

Geographic locations may be assigned to variables using geo URL syntax, followed by an optional label. For example Los Angeles is defined as:

la="geo:34.05,-118.25<tab>Los Angeles"

The geographic bounding box of a map is defined by latitude (-90° to 90°) and longitude (-180° to 180°) in decimal degrees.

The 'geobound' keyword maps the specified lat/long boundaries to the slide canvas boundaries set with the 'geocanvas' keyword. The defaults for geobound are -90 90 -180 180, (min,max latitude and min,max longitude) with geocanvas defaulting to 0 100 0 100 (min,max X and min,max Y).


```
deck
    slide
        geobound -34.83 37.34 -17.52 51.266 // boundary of Africa
        geocanvas 15 85
        cairo="geo:30.04440,31.235700    Cairo"
        accra="geo:5.614818,-0.205874    Accra"
        geoloc    cairo "b" 4
        geoloc    accra "e" 4
        georegion "africa.kml" "tan"
        geopath   accra cairo 0.5 "red"
    eslide
edeck
```

A map shows the way.